Predicting Hard Drive Failure from Fleet Telemetry

IST 707 Machine Learning · Midterm Presentation · October 20, 2026

Tyler Katz · Hashim Khan · Mrgaj Iyer

310.9M drive-days · 72 drive models · 2023 to 2025

The Problem: Drive Replacement Is Reactive

- Data centers run hundreds of thousands of spinning drives. Failures are rare per drive, but at fleet scale a dozen happen every day.
- Today a drive is replaced when a firmware warning counter trips. Only about half of failing drives ever show one.
- Operators work with a fixed weekly budget of spares and technician hours.

Our goal: predict failure in the next 30 days from the last 30 days of telemetry, and rank every drive by risk so a fixed budget replaces the right ones first.

1.3%

of fleet drives fail per year (Backblaze, 2025)

47.9%

of failing drives showed bad sectors in the prior 30 days

30 days

the warning window our model predicts

Stakeholders and Why It Matters

1

Cloud storage operators

Backblaze splits every file into 20 pieces with 3 spares. Each failed drive burns safety margin until rebuilt. A ranked list turns emergency swaps into planned work within a fixed budget.

2

Enterprise IT teams

A 500-drive shop sees 6 or 7 failures a year: far too few to learn patterns from its own data. They can borrow the patterns learned from a fleet of hundreds of thousands.

3

Drive manufacturers

SMART reporting is inconsistent: the same temperature reads 32, 100, or 59 depending on vendor. Evidence of which readings predict failure supports standardization.

Each stakeholder needs something different from a prediction: planned replacement, early warning, or evidence about the sensors themselves.

Related Work: Thresholds Have a Known Ceiling

Current practice

Firmware thresholds. Backblaze (2016): 76.7% of failed drives tripped a watched counter, so 23.3% gave no warning at all.

2007 field studies

Google (Pinheiro et al.): over 56% of failed drives had zero count on the four strongest signals. Schroeder and Gibson: real replacement rates of 2 to 4% per year, far above datasheets.

ML, 2005 to 2021

SVMs beat the built-in alarm (Murray 2005); trees caught 95% of failures (Li 2014); Aussel 2017 used this same dataset; Lu 2020 needed extra non-SMART signals to go further.

What we add

- Test on a year the model never sees (train 2023 to 24, test 2025)
- Metrics an operator uses: precision and recall at a fixed replacement budget
- Six models compared across three years of data

Nine sources cited in the report. SMART-only data caps recall for us too: that limit is inherited, not solved.

Dataset: Backblaze Drive Stats, 2023 to 2025

314.5M

raw drive-days across 12 quarters

72

hard drive models after removing 22 SSD models

89

features per 30-day window (SMART first, last, change)

3.99M

drive-window rows in the modeling table

Split	Period	Rows	Failures	Rate
Train	Jan 2023 to Oct 2024	169,827	8,087	sampled 20:1
Calibration	Nov to Dec 2024	297,009	328	1 in 905 (true)
Test (held out)	Jan to Dec 2025	3,521,973	3,722	1 in 946 (true)

- Chronological split: the model never sees the year it is judged on
- Labels look one window ahead; windows at split boundaries are dropped
- Cross-checked against Backblaze's published totals: within 0.5%

Why we trust it: collected automatically by the operator that replaces the drives, published since 2013, used in prior research.

How the Modeling Table Was Built

1 Raw archives

314.5M drive-days in 12 quarterly zips, one CSV per day

2 Narrow the columns

179 to 33 columns; 22 SSD models dropped

3 Reshuffle by drive

64 buckets by a hash of serial number, so a drive's history sits together

4 30-day windows

per sensor: first, last and change; 84 SMART + 5 descriptors = 89 features

5 Label from the next window

label = failure in the NEXT 30 days; the only value that looks forward

6 Split by calendar

train 2023 to Oct 2024, calib Nov to Dec 2024, test 2025; boundary windows dropped

7 Sample negatives

train only: 20 healthy per failure, fixed seed; calib and test untouched

8 The modeling table

three Parquet files, 94 columns; deterministic, rebuilds exactly

314.5 million drive-days in, 3.99 million drive-window rows out, with no feature allowed to see the future and no split allowed to see another.

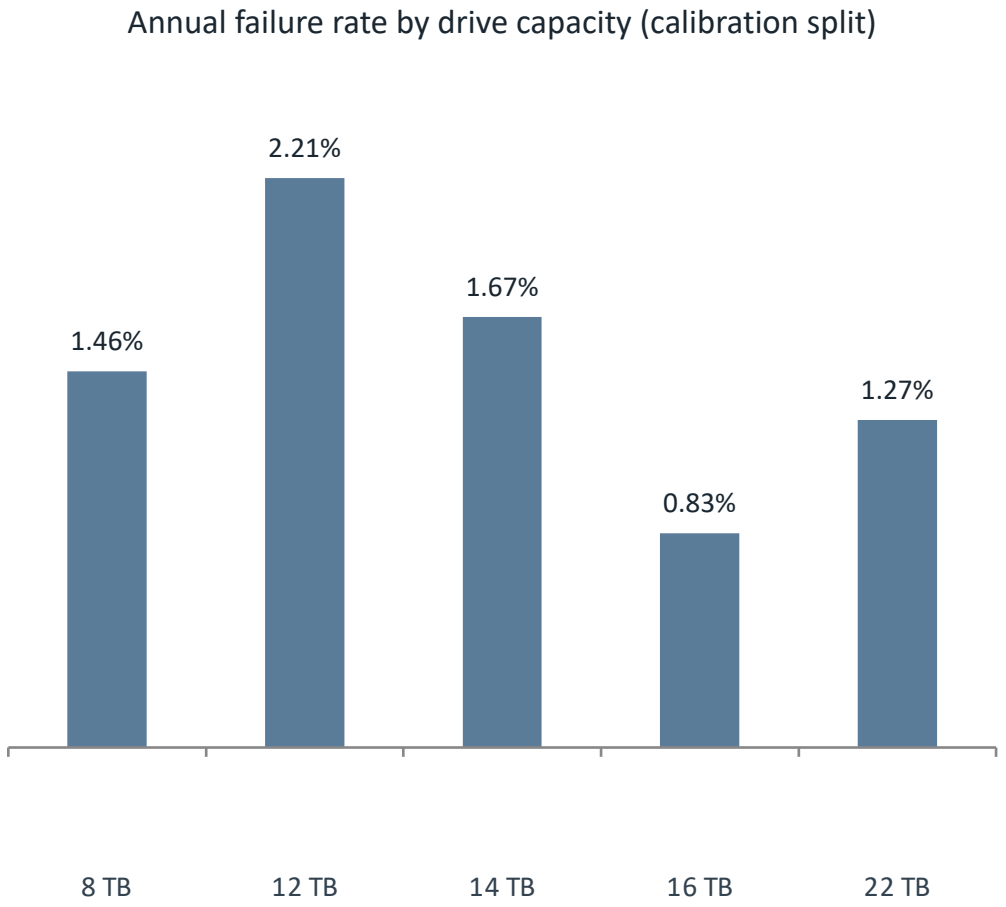
EDA 1 and 2: Imbalance, and Who Fails

Severe class imbalance (Analysis 1)

- True rate: 1 failure per 905 windows (calibration split)
- Predicting "healthy" for everything scores 99.9% accuracy, so accuracy is out
- Train is sampled 20:1, which is 43x the real rate: probabilities must be recalibrated

Failure depends on who built the drive (Analysis 2)

- HGST fails at 2.61% per year, about 3x WDC's 0.85%
- Age matters: 0.92% in year 1, lowest at 1 to 2 years (0.68%), peaking at 3.56% at 5 to 6 years
- Chi-square rejects independence from model (p near 1e-17), but the effect is small (Cramer's V 0.020)



EDA 3: Three Counters Carry the Signal

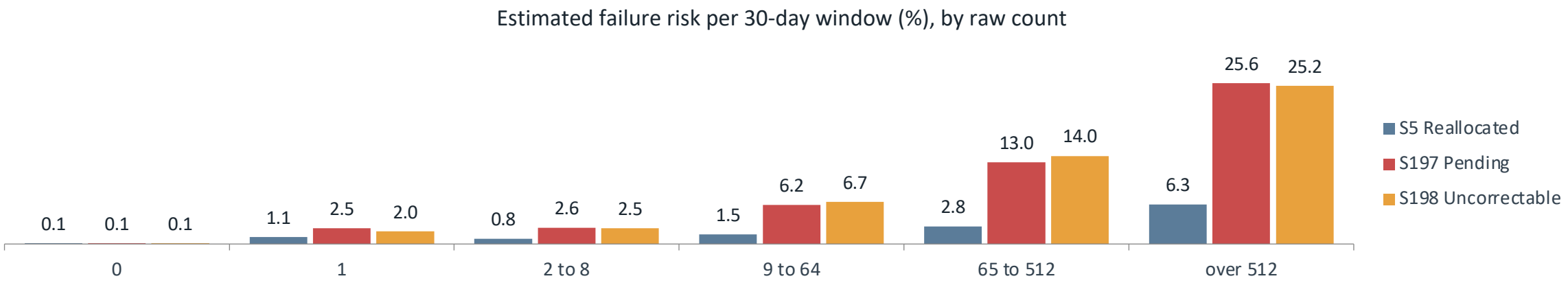
Any non-zero reading is a red flag (Analysis 3)

Attribute	Non-zero, healthy	Non-zero, failed	Odds ratio
S5 Reallocated sectors	3.4%	47.9%	25.9
S197 Pending sectors	1.2%	41.5%	59.0
S198 Offline uncorrectable	0.8%	30.9%	55.9

But about half of failing drives show nothing in S5: sudden electronic and mechanical failures cap the recall SMART alone can reach.

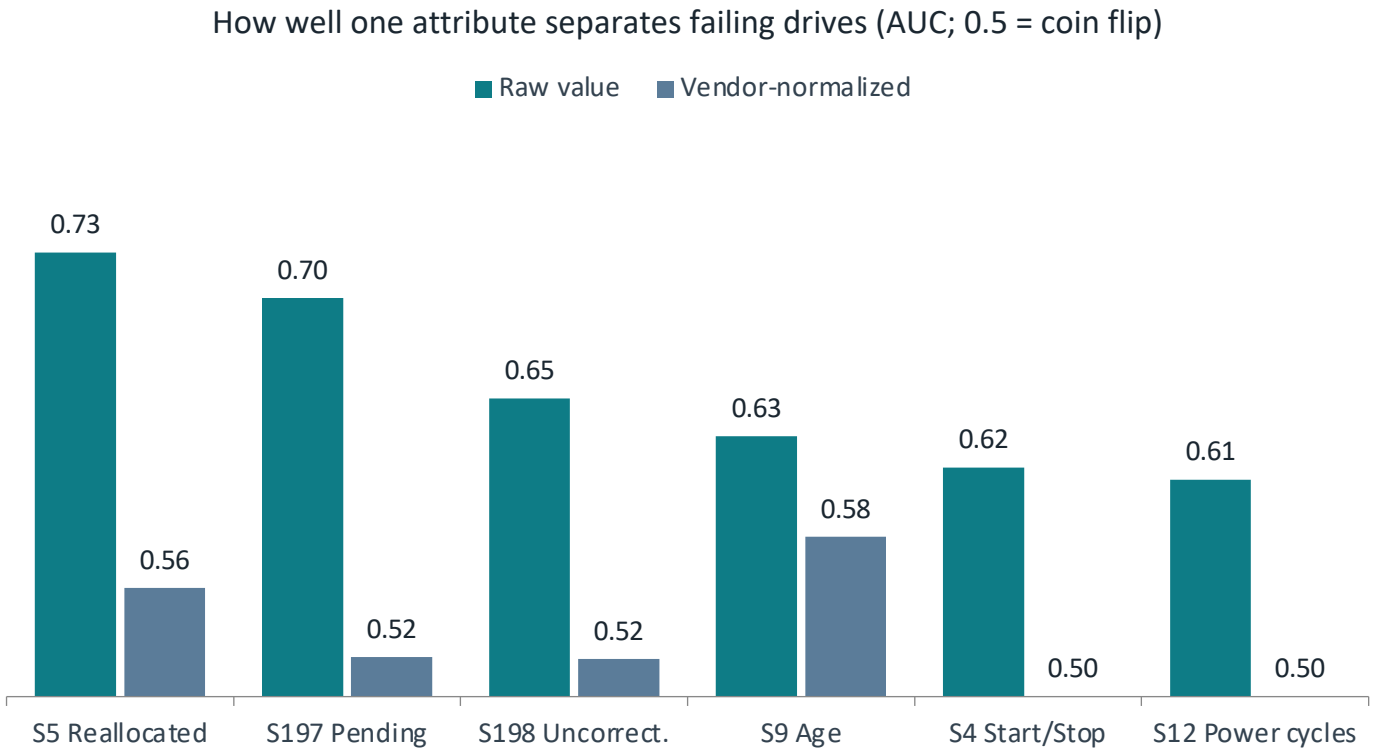
The EDA's verdict

The signal is real but partial. S5, S197 and S198 raise failure odds 26x to 59x, yet about half of failing drives show no reallocated sectors. That gap is exactly what a multi-feature model is for.



EDA 4: Raw Values Beat Vendor-Normalized Ones

Raw counters predict failure better (Analysis 4)



Normalized values squeeze up to 72,495 raw values into as few as 48 levels and stay flat until damage is severe. Across all 14 attributes raw wins (Wilcoxon p = 0.009).

One temperature, four encodings

Vendor	Normalized value
Seagate	32 (the real °C)
Toshiba	100 (always)
WDC	59
HGST	176

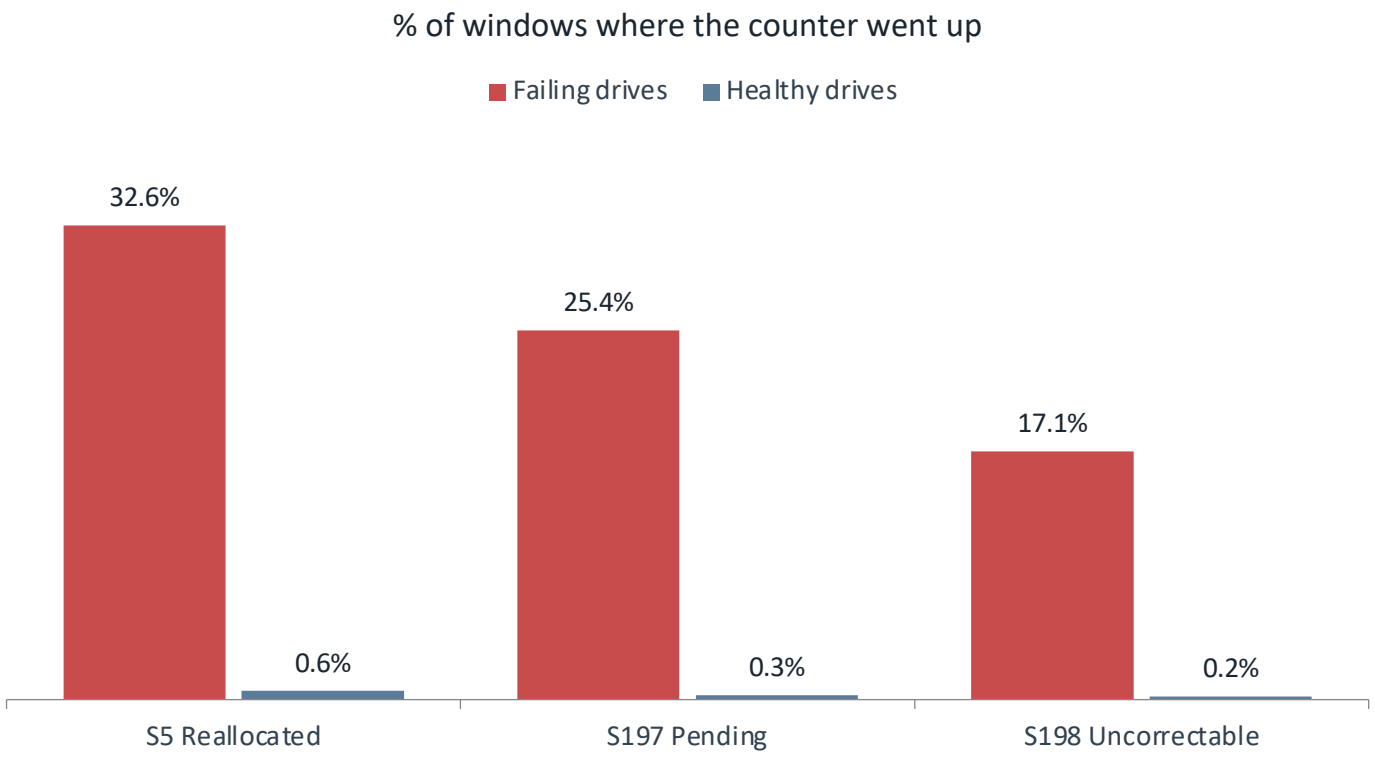
Healthy drives at about the same real temperature (32 to 35 °C).

What we do

Train on raw values, keep vendor as its own feature, and drop normalized columns that duplicate raw values or are confounded with vendor.

EDA 5: Warning Signs Grow Before Failure

Counters that rose during the final 30 days (Analysis 5)



A rising counter is 54x (S5), 82x (S197) and 97x (S198) more common in failing drives. Age (S9) rises in 99.9% of windows for both, so its change only measures time.

Redundant pairs (Spearman ρ)

Pair	ρ	What we do
S4 ↔ S12	0.997	Keep one
S197 ↔ S198	0.80	Keep both
S1 ↔ S7	0.90	Vendor artefact
Age ↔ capacity	−0.85	Confounded

Bigger drives are newer, so age and capacity move together.

What we do

Keep both the end-of-window level and the 30-day change for S5, S197 and S198. Drop one of S4 and S12.

Baselines: The Floor Any Model Must Beat

Operational rule (no training)

Flag any drive with reallocated sectors (SMART 5 > 0): the counter operators watch today.

50.3%

of next-month failures caught

1 in 75

flagged drives actually fails (1.3% precision)

It flags 42 of every 1,000 drives and gives a yes or no, not a ranking: the operator cannot choose how many to replace.

Random baseline (null hypothesis)

Flag drives at random: what "no information" achieves on this data.

0.11%

precision at any budget (the failure rate)

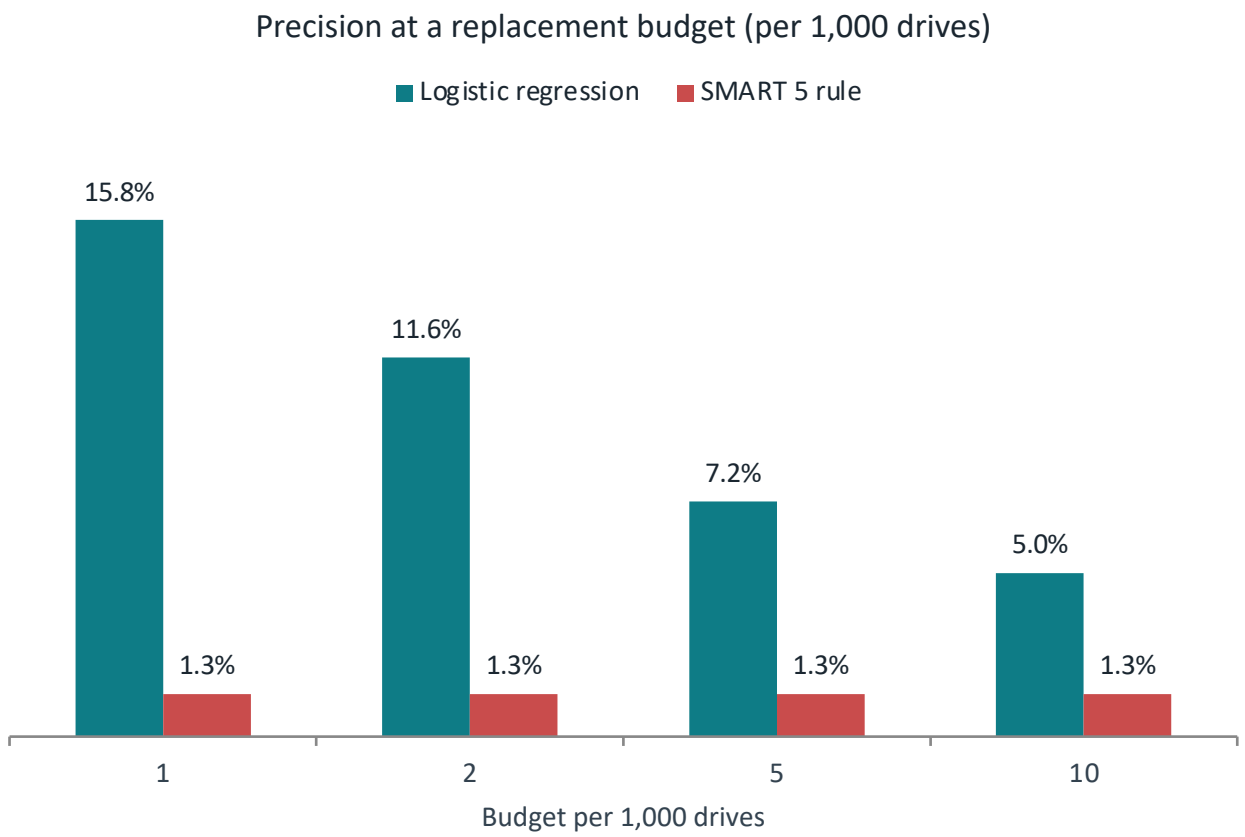
0.0011

PR-AUC floor

The rule is about 12x better than random at picking failing drives, but almost every drive it flags is healthy.

Both scored on the calibration split at the true failure rate (1 in 905). Metrics: precision and recall at a fixed replacement budget, plus PR-AUC.

Best Model So Far: Logistic Regression



Scored on the calibration split at the true failure rate. At the rule's own budget (42 per 1,000) the model catches 66.2% of failures vs 50.3%. The 2025 test year remains untouched.

11x

the rule's PR-AUC: 0.082 vs 0.0073 (random 0.0011)

1 in 6

flagged drives fails at a 1-per-1,000 budget, vs 1 in 75 for the rule

Calibrated honestly: threshold and Platt scaling chosen on one half of calibration, checked on the other. Predicted risk went from 28x reality to 1.1x.

Future Work: Six Models, One Honest Test

Logistic regression
done: the floor-setter

Decision tree
readable rules

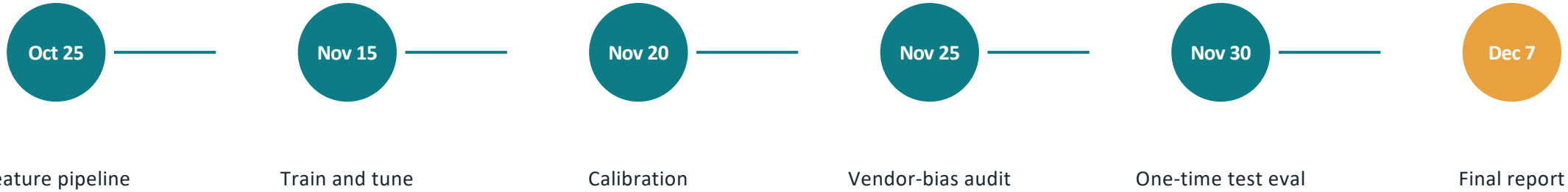
Random forest
stability

XGBoost
strongest on tables

SVM
a different boundary

LightGBM
scale + categories

All six share the same EDA-chosen inputs: S5, S197, S198 raw values and changes, age, wear, temperature, vendor.



The test split is opened once, after models and calibration are locked. The manufacturer-bias audit checks the model reads drive health, not the brand.

Problems and Challenges, and the Workarounds

Class imbalance (1 in 905)

Budget-based precision and recall instead of accuracy; a calibration split at the true rate; Wilson and Poisson intervals for rare counts.

Downsampled training data

Each kept healthy window stands for about 36 real ones: rates reweighted, probabilities recalibrated with Platt scaling.

Vendor-dependent sensors

S197 missing only on two Toshiba models: kept as missing plus an indicator. Raw values used; vendor kept as an explicit feature.

Fleet shift over time

Only 9 of 72 drive models are unseen in training, but 67% of 2025 test drives never appear in it (mostly from down-sampling): we encode vendor and capacity rather than drive-model identity.

Every workaround is already in the pipeline or the notebooks, not on a wish list.

A Ranked List Worth Acting On

One model in, the approach already beats the tool operators use today: 11x the rule's PR-AUC, and 1 in 6 flagged drives failing where the rule manages 1 in 75.

Five models, a bias audit, and one untouched test year still ahead.

Questions welcome